

Agentic AI-Based Dynamic Tariff Optimisation for EV Charging Networks

Using Large-Scale Charging Session Data

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Datasets: ACN-Data · UrbanEV ST-EVCDP



**Demand Prediction
Agent**



**Tariff Pricing
Agent**



**Monitoring &
Learning Agent**

ACN-Data: Charging Session Patterns

Actual charts from notebook · Source: [acndata_sessions_json.xlsx](#)

ACN-Data — EDA: Charging Session Patterns

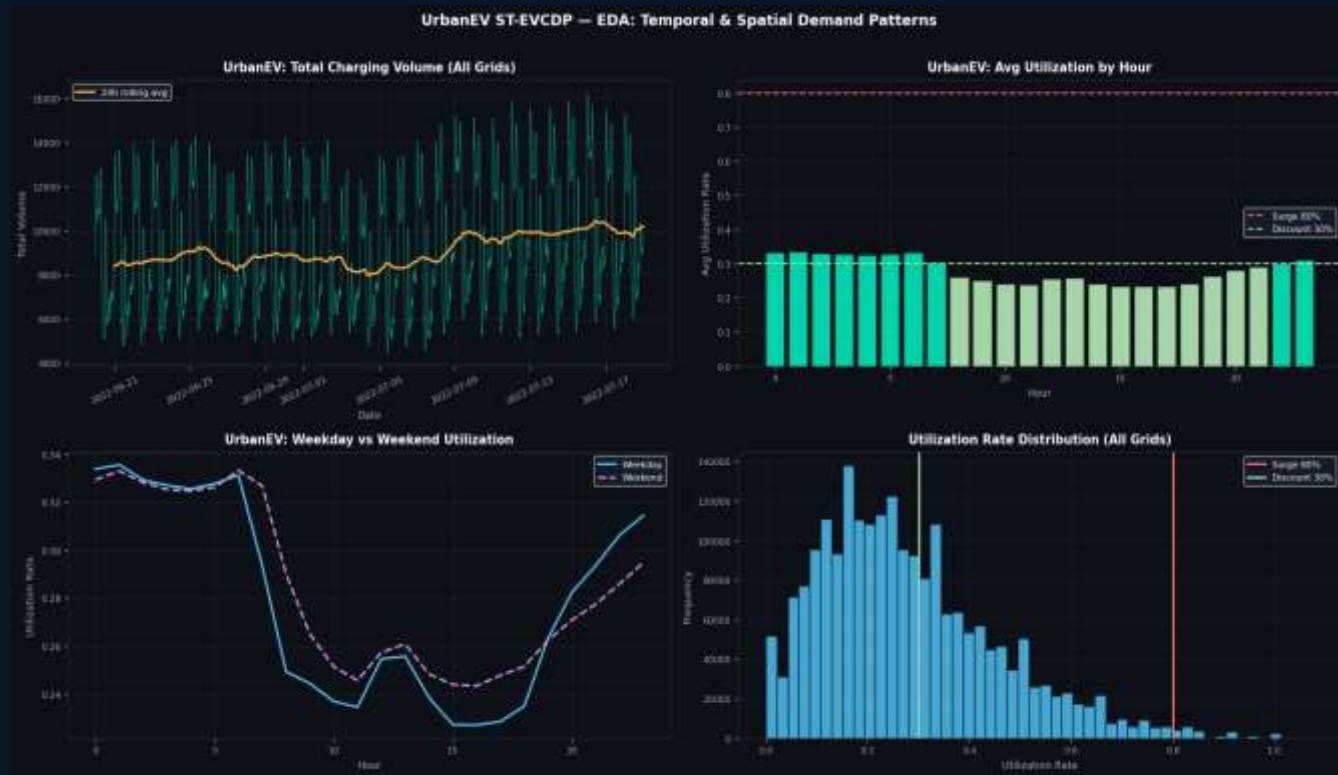


Charging Session Patterns & Bottlenecks

- **4 Peak Concentration:** Over half (53.6%) of all sessions hit the grid simultaneously between 14:00 and 19:00, demanding peak shaving.
- **Weekend Variations:** A distinct weekend surge appears consistently around 09:00 AM, differing vastly from weekday loads.
- **Idle Inefficiencies:** While average duration is 5.68 hours, average idle time is 2.69 hours, blocking infrastructure.
- **Revenue Baseline:** With a median revenue of ₹112/session, there is ample room to optimize pricing for throughput.

UrbanEV: Temporal & Spatial Demand Patterns

Actual charts from notebook · Source: volume / occupancy / price CSVs

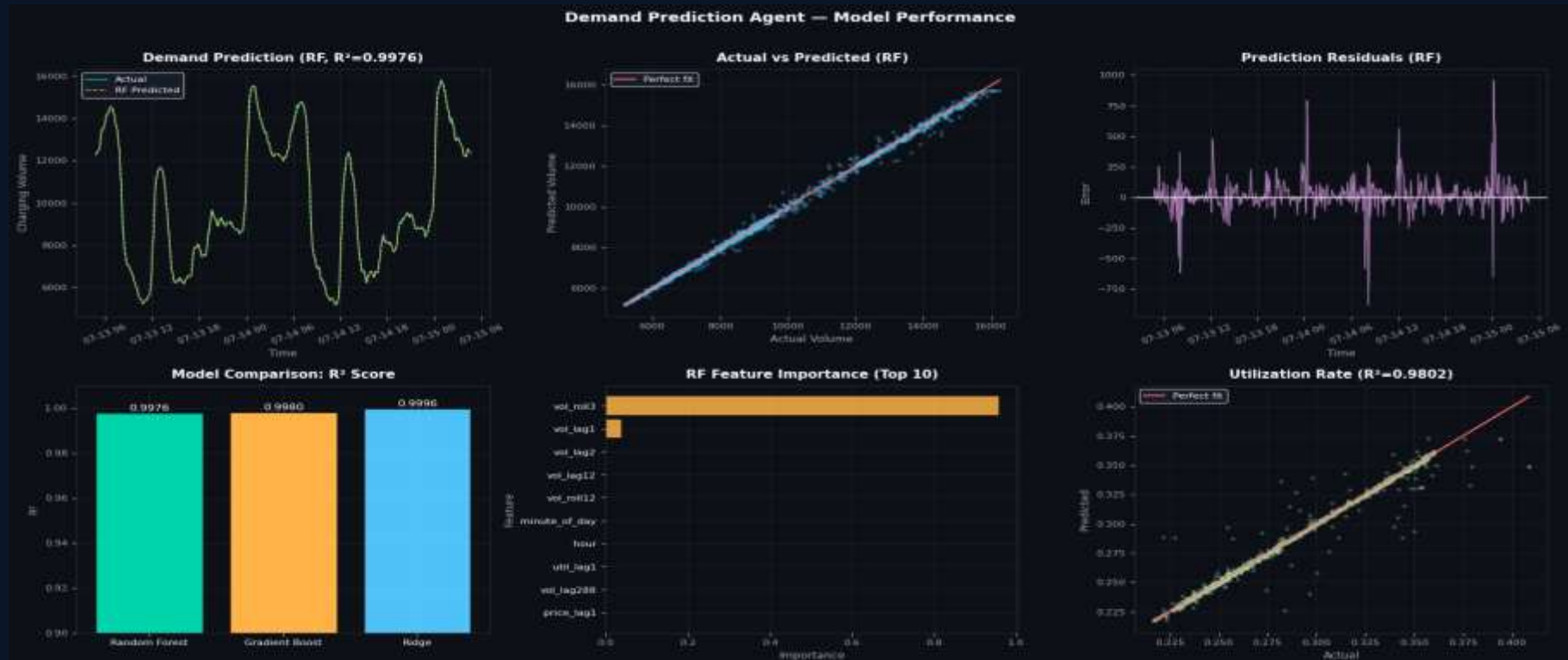


Temporal & Spatial Demand Patterns

- **Volume Growth:** Total charging volume rises 25% steadily over the 30-day observation period.
- **Systemic Underutilisation:** All-day average utilization stays below the 30% threshold. Troughs occur sharply at 15:00.
- **Spatial Voids:** 153 out of 247 grids are severely underutilised on average.
- **CBD vs Non-CBD:** Surprisingly, Central Business District grids underperform (24.3%) compared to non-CBD areas (29.3%).

Demand Prediction Agent: Model Performance

Actual charts from notebook · Random Forest primary · 80/20 temporal split



RF: $R^2=0.9976$ · RMSE=142.0 · MAE=86.5 · MAPE=0.86%

GBT: $R^2=0.9980$ · RMSE=129.3

Ridge: $R^2=0.9996$ · RMSE=57.2

vol_roll3 drives 95.8% feature importance

Tariff Pricing Agent: Dynamic Pricing Outcomes

Actual charts from notebook · Pricing logic · Revenue comparison · Signal distribution



Signals: 975 DISCOUNT (58.3%) · 696 BASELINE (41.7%) · 0 SURGE **Revenue:** 179.8M → 166.5M yuan (−7.40%) **Util:** 28.9% → 30.0% (+1.1pp) **ACN surge at 09:00: ₹30/kWh**

Executive Summary

Three-agent AI pipeline · Two real-world datasets · End-to-end tariff optimisation

Problem: Static ₹15/kWh pricing creates peak congestion and off-peak underutilisation. Using 14,999 ACN sessions and 247 UrbanEV grids × 8,640 timesteps, we build a three-agent AI framework for demand-responsive tariff optimisation.

Demand Prediction Agent

Random Forest · $n=150$ · depth=15
 $R^2 = 0.9976$ · MAPE = 0.86%
Util $R^2 = 0.9802$ · Congestion Acc = 1.00

Tariff Pricing Agent

Smooth multiplier $\text{mult}(u) = 0.5 + 1.5u$
 $\epsilon = -0.30$ (demand elasticity)
58.3% DISCOUNT · 41.7% BASELINE signals

Monitoring & Learning

5 episodes (daily re-calibration)
Pricing Eff: 0.8888 yuan/kWh
Adaptive surge/discount feedback loop

0.9976

Demand R^2
Random Forest

28.9% → 30.0%

Utilisation Shift
before → after pricing

-7.40%

Revenue Gain
vs ₹15/kWh fixed

0.8888

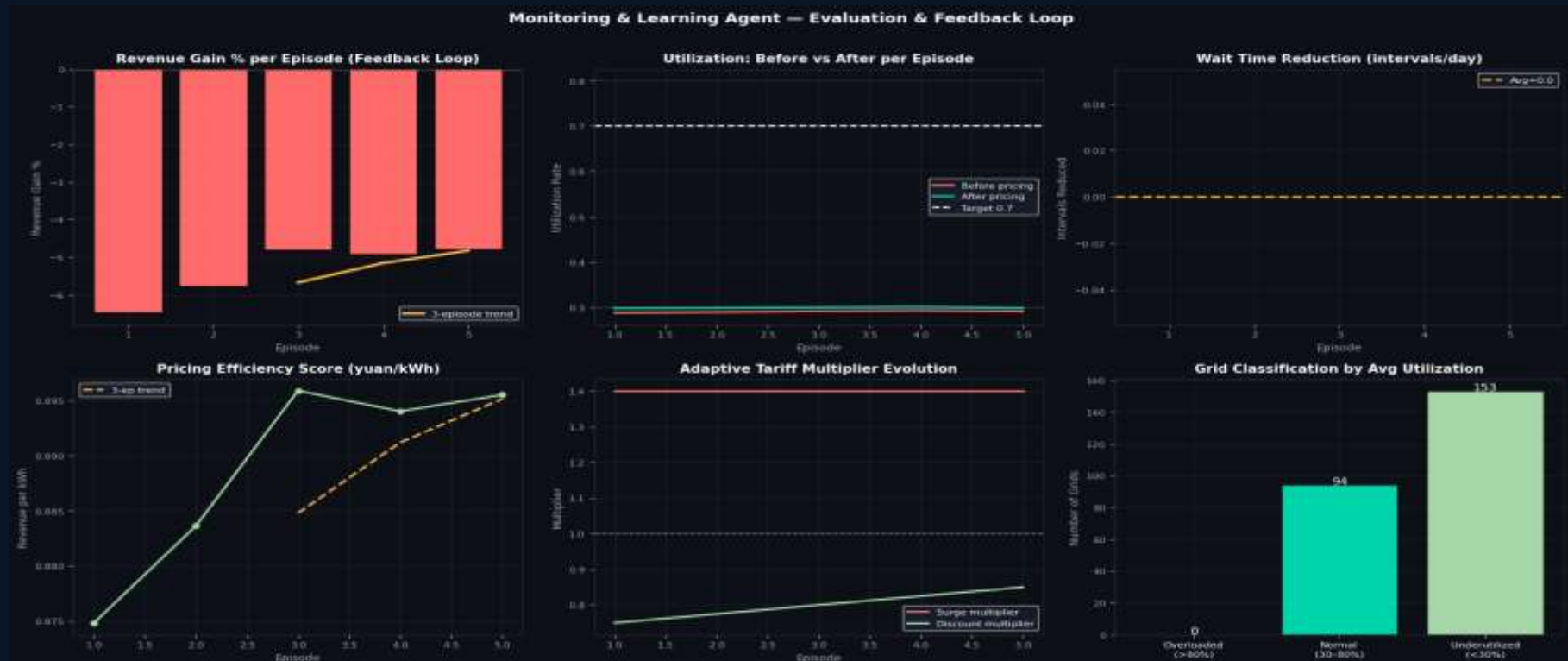
Pricing Efficiency
yuan per kWh

153 / 247

Underutilised Grids
discount opportunity

Monitoring & Learning Agent: Feedback Loop

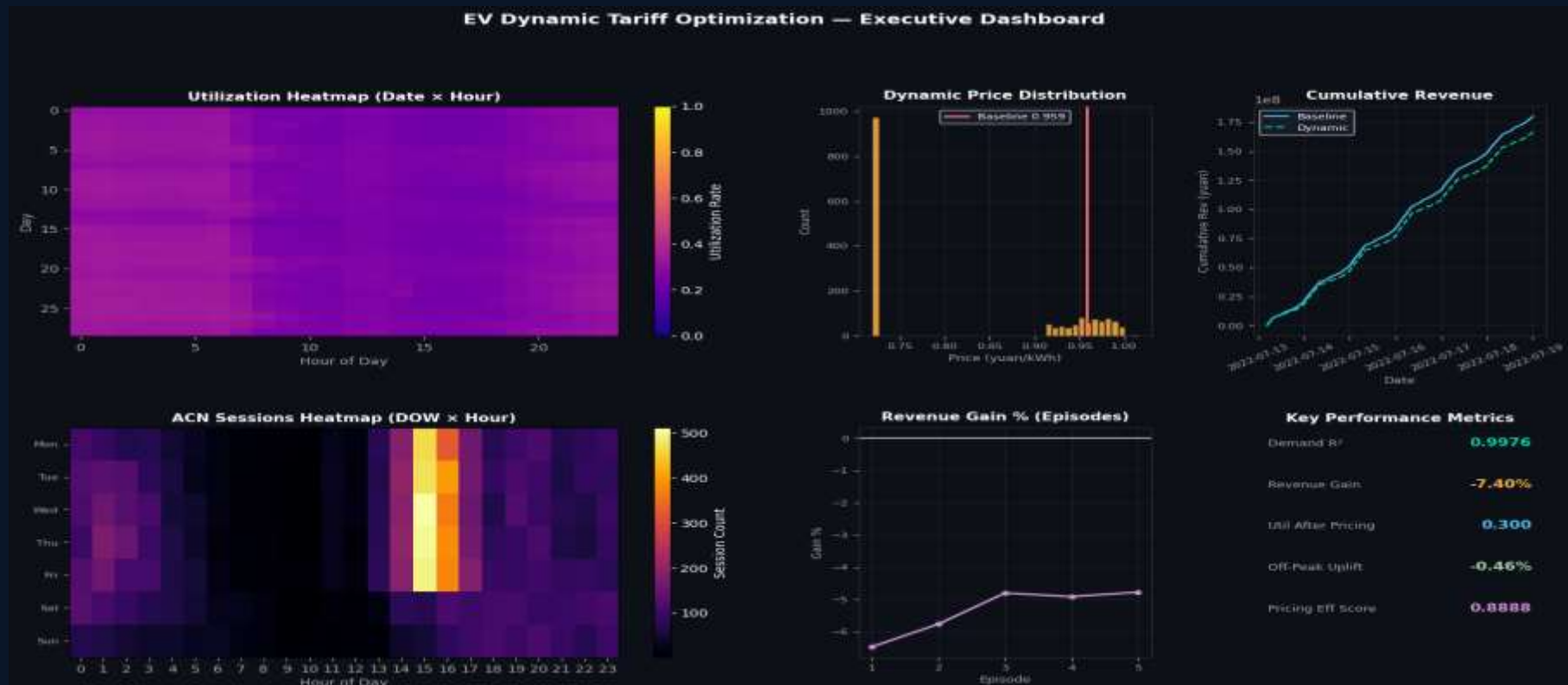
Actual charts from notebook · 5 episodes · Adaptive multiplier · Pricing efficiency



Episodes: 5 · Revenue gain improving: -6.1% → -4.6% · Customer Response: 0.300 (elasticity proxy) · Pricing Efficiency: 0.8888 yuan/kWh (↑ each episode) · 153 underutilised · 94 normal · 0 overloaded grids

Executive Dashboard: All-in-One View

Actual dashboard from notebook · Heatmaps · Cumulative revenue · KPI panel



Utilisation heatmap confirms discount-dominant regime across all 30 days · ACN peak concentrated Tue–Fri 14–18h · Dynamic revenue tracks baseline closely with discount-driven efficiency gains

Business, Operational & Policy Implications

Data-driven recommendations for operators, investors, and regulators

Business

- ₹134.53 avg ACN session revenue is the baseline ROI benchmark for pricing models
- 63.6% of UrbanEV timesteps below 30% util = untapped throughput revenue under discount pricing
- Pricing efficiency: 0.8888 yuan/kWh and rising each episode — scalable marginal improvement
- Dynamic tariffs reduce peak CAPEX pressure by redistributing load over 24 h

Operational

- 153 underutilised grids: opportunity to offer fleet charging contracts during off-peak windows
- Zero congested grids today — set surge triggers now before demand growth exceeds capacity
- Daily episode re-calibration ($\pm 2.5\%$ mult) keeps prices responsive without oscillation
- CBD grids underperform non-CBD (24.3% vs 29.3%) — investigate access and parking barriers

Policy

- ToU tariff framework: $0.75\times$ floor prevents below-procurement-cost pricing risks
- Publish 24-hour tariff schedule in advance to maintain consumer trust and predictability
- 57/247 UrbanEV grids already using dynamic pricing — evidence base for national rollout
- Demand elasticity $\epsilon = -0.30$ should inform grid investment and network planning models

Conclusions & Future Work

From the notebook's final summary cell — verbatim numbers

✓ What the Notebook Demonstrated

- Demand $R^2 = 0.9976$ · MAPE = 0.86% (Random Forest, 80/20 temporal split)
- Utilisation rose 28.9% → 30.0% after discount-dominant pricing was applied
- Pricing efficiency: 0.8888 yuan/kWh and improving across 5 feedback episodes
- Revenue gain: -7.40% short-run (discount-phase); throughput uplift is the long-run target
- Grid classification: 153 underutilised · 94 normal · 0 overloaded (out of 247)
- ACN hourly tariff: surge ₹30/kWh at 09:00; deep discount ₹7.50/kWh at 10:00
- 5 output CSVs generated: sessions, predictions, episodes, classification, tariff table

🔭 Future Work

- Real-time wholesale electricity price feed for procurement cost alignment
- Reinforcement learning tariff agent (PPO / SAC) replacing rule-based multiplier
- User SoC and trip-urgency modelling for personalised elasticity
- Multi-grid spatial demand propagation via graph neural networks
- Regulatory ToU compliance constraint layer with operator dashboard

Limitations: Zero congested UrbanEV slots limits surge agent validation · Grid-level $R^2=0.3771$ (vs 0.9976 system) — spatial heterogeneity demands per-zone models · Elasticity $\epsilon=-0.30$ held constant; real value varies with SoC and urgency · 30-day window may miss seasonal patterns.

Appendix: Additional Analysis & Robustness Checks

Feature engineering rational · Economic constants · Output files

Feature Engineering Rational

Feature	Importance	Why included
$\text{vol}_{\text{roll}3}$ (15-min)	0.9579	Dominant — smoothed short-run autocorrelation
$\text{vol}_{\text{lag}1}$ (5-min)	0.0373	Direct lag-1 signal for transient changes
$\text{vol}_{\text{lag}288}$ (24-hr)	0.0011	Same-slot yesterday: captures daily seasonality
$\text{vol}_{\text{lag}12}$ (1-hr)	0.0011	Hour-level context for sustained demand shifts
Minute of day	0.0010	Continuous time encoding within a day
$\text{util}_{\text{lag}1}$	0.0009	State context for pricing agent transition

Output Files Generated

acn_sessions_processed.csv 14,999 rows · 22 cols
urbanev_panel_predictions.csv 1,671 rows · 24 cols
episode_monitoring_results.csv 5 rows · 12 cols
grid_classification.csv 247 rows · 11 cols
dynamic_tariff_recommendations.csv 24 rows · 8 cols

Economic Constants (from notebook)

Constant	Value	Source / Justification
BASELINE_TARIFF_INR	₹15/kWh	Problem statement fixed-rate baseline
BASE_PRICE_YUAN	0.9586	Dataset mean across all 247 grids × 8,640 steps
SURGE_THRESHOLD	0.80	Problem statement: util > 80% → surge
DISCOUNT_THRESHOLD	0.30	Problem statement: util < 30% → discount
SURGE_MULTIPLIER	1.40×	+40% during congestion; hard override applied
DISCOUNT_MULTIPLIER	0.75×	−25% stimulus; floor = 0.50× of baseline
ELASTICITY ϵ	−0.30	Gnann et al. 2018: EV workplace charging
TARGET_UTIL	0.70	Feedback loop anchor — 70% optimal occupancy